Dividing polynomials

1 Simplify the following:

a $\dfrac{15k}{20k}$ b $\dfrac{70v}{710}$ c $\dfrac{42xy}{18y}$ d $\dfrac{18p^7}{9p^3}$

e $\dfrac{30mn^2}{9m^2n}$

2 Consider the following statement $x^9 \div x^3 = x^3$ for all nonzero real numbers x.

a Determine whether the statement is true or false.

b FInd $x^9 \div x^3$.

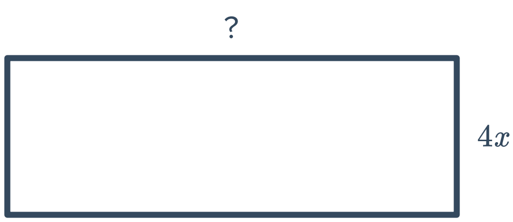
3 Simplify the following:

a $\dfrac{z+10}{10}$ b $\dfrac{5x^3+4x^5}{x}$

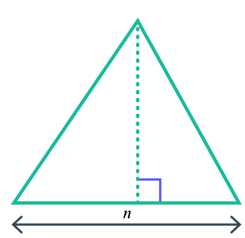
c $\dfrac{18x^3-12x^2+24}{3}$ d $\dfrac{20x^4-35x^3+15x^2+30x}{5x}$

4 Consider the area of the following rectangles.
Find a polynomial expression for its length.

a Area $= \left(4x^4 - 8x\right)$ square units b Area $= \left(6x^3+4x^2+10x+14\right)$ square units



5 The triangle shown below has an area of $13n^3 + 11n^2 + 29n$.
Find a polynomial expression for its height.



6 Factorise each of the following and simplify.

a $\dfrac{6x-16}{12}$

b $\dfrac{x-4}{4-x}$

c $\dfrac{3y-45}{15-y}$

d $\dfrac{m^2-100}{m-10}$

e $\dfrac{m^2-1}{1+m}$

f $\dfrac{a^2-81}{9-a}$

g $\dfrac{x^2+4x+4}{10(x+2)}$

h $\dfrac{x^2+8x+16}{(x-5)(x+4)}$

i $\dfrac{x^2+7x+10}{x+2}$

j $\dfrac{2x^2+4x-198}{5(x-9)}$

k $\dfrac{2x+8}{x^2+4x}$

l $\dfrac{20-5x}{2x^2-8x}$

m $\dfrac{m^2-1}{2m+2}$

n $\dfrac{2x^2+10x-100}{2x+20}$

o $\dfrac{8x^2+2x-15}{(4x-5)^2}$

7 At a landfill site, a hole in the shape of a rectangular prism is to be dug out. The area of the rectangular cross-section measures $(5y+8)$ square metres and the volume of the hole is to be $5y^2+3y-8$ cubic metres such that $(5y+8) \times \text{depth} = 5y^2+3y-8$. Find the expression for the depth of the hole.

Additional questions

8 Fill in the blanks to make a true algebraic statement.

a $\dfrac{\square}{5ab} = 6bc$

b $\dfrac{2x^3-10x^2+8x}{\square} = x^2-5x+4$

c $\dfrac{\square+20m^2n^2+21m^3n}{14\square} = n^2+\dfrac{10}{7}mn+\square$

9 Consider the following statement: $x^9 \div x^3 = x^3$ for all nonzero real numbers x

a What is $x^9 \div x^3$ actually equal to?

b Identify a nonzero real number for which the original statement is *true*.

10 Consider the problem $\dfrac{24x^6}{6x^2}$.

Identify and correct the error in each of the following student's work.

Lawrence:

$\dfrac{24x^6}{6x^2} = 18x^4$

Marika:

$\dfrac{24x^6}{6x^2} = 4x^3$

11 Determine whether each statement regarding the division of polynomials by monomials is true or false. Justify your conclusion.

- **a** When dividing a polynomial by a monomial, it is possible to reduce the number of terms.
- **b** The degree of the polynomial will always decrease after dividing by a monomial
- **c** The result of dividing a polynomial by a monomial will have a constant term if the degree of the monomial matches the degree of any term in the polynomial.
- **d** The result of dividing a polynomial by a monomial will be another polynomial.

12 Explain the difference between the two expressions:

$$\frac{5x^2+10x+5}{5} \qquad\qquad 5x^2+10x+5/5$$

13 Create an example and describe the result when a polynomial is divided by a monomial that:

- **a** Has a coefficient larger than the coefficients of the terms in the polynomial.
- **b** Has a variable with a larger degree than the terms in the polynomial.